The Blessings of Overparameterization

Applications in Solving Economic Models

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The moral
The parameter count was never the risk.

Motivation

- Deep learning solves **high-dimensional** dynamic models once thought intractable.
- The networks are **massively overparameterized**: far more parameters than "data" points.
- Classical worry: **"overfitting"** out of grid.
- Difference: residuals are deterministic, so **no noise to overfit**. What matters is **between the grid points**.

The question

When we solve dynamic models with neural networks, is overparameterization a **necessary evil**, or does it buy us something?

Setup: models as functional equations

An economic model is an equation (Euler or Bellman):

$$\ell(x, f^\star) \quad = \quad \mathbf{0}$$

Fit $f(\cdot;\theta)$ by minimizing the residual loss

$$\mathcal{L}(\theta;\mathcal{D}) \quad = \quad \frac{1}{N}\sum_{x\in\mathcal{D}} \big\|\ell\big(x, f(\cdot;\theta)\big)\big\|_2^2$$

Neural nets: **overparameterized** ($M \gg N$), **non-convex**. Solution depends on **initialization**.

What we find

Empirically
- **Accuracy improves with size.**
- **Stability improves with size:** the seed spread collapses.

Theoretically
- **Width buys agreement; the implicit bias decides what you agree on.**

Four models of increasing difficulty

- **Linear-Quadratic:** closed form, sharpest test.
- **McCall:** 1D, kink at the reservation wage.
- **RBC:** 2D stochastic, no closed form (vs. VFI).
- **Small Open Economy:** 5D, beyond any grid.

Here: the first two, plus the random-features dissection.

The two worked models

Linear-Quadratic: firm chooses investment u, quadratic adjustment cost, state (y, Y).

$$v(y,Y) = \max_u \ (\alpha_0 - \alpha_1 Y)y - \tfrac{\gamma}{2}u^2 + \beta\, v(y', Y')$$
$$\text{s.t.} \quad y' = y + u, \qquad Y' = h_0 + h_1 Y$$

McCall: worker accepts offer $w \sim f$ on $[0, B]$ or keeps searching.

$$v(w) \quad = \quad \max\left\{\tfrac{w}{1-\beta},\ c + \beta\int_0^B v(w')f(w')\,dw'\right\}$$

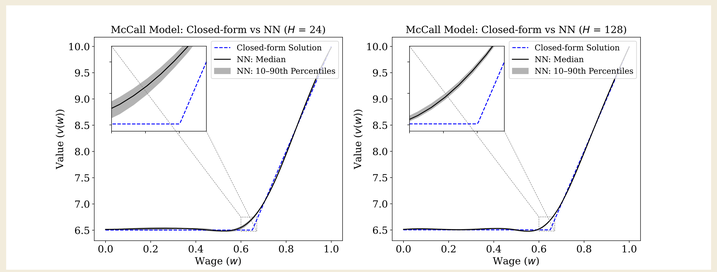
Result 1 – Linear-Quadratic regulator

Closed form, so error is exact. Euler residual on $n_Y = 6$ points; train and test loss **indistinguishable** at every width.

Median and max error fall to $\sim 10^{-4}$. The **seed band narrows**.

Result 2 – McCall job search

Non-smooth: a **kink** at the reservation wage, learned from only $N = 11$ points.



Narrow ($H = 24$): seeds disagree near the kink. Wide ($H = 128$): all 50 **converge to the same function**. **Stability confirmed.**

A careful examination: random features

Freeze the first layer, train only the outer weights v:

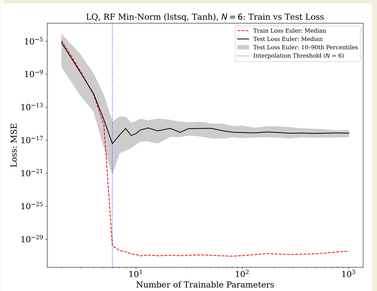
$$f_H(Y;\theta) \quad = \quad \sum_j v_j\,\sigma(\tilde{w}_j Y \ + \ \tilde{b}_j) \ + \ c$$

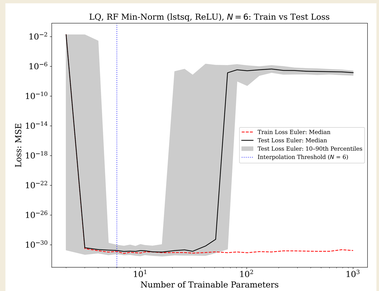
The LQ residual is **linear in** θ: $\Phi\theta = d$ with $H + 1 > N$, so **infinitely many solutions**. **The algorithm is the identifying assumption**. Gradient descent from zero picks

$$\theta^\star \quad = \quad \underset{\theta:\,\Phi\theta=d}{\operatorname{argmin}} \ \|\theta\|_2 \quad = \quad \Phi^\top(\Phi\Phi^\top)^{-1}d,$$

computed directly, **bypassing training**.

Minimum norm knows no economic model

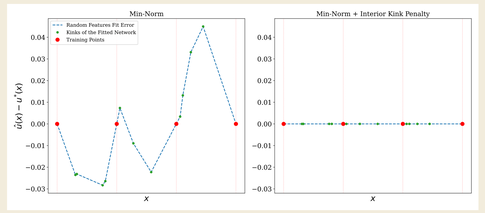




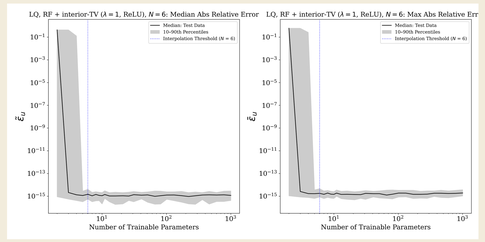
Tanh works: test loss stays flat. **ReLU fails:** test loss rises with width.

Train loss is zero for both; min-norm in **coefficients**, not in **functions**:
- **Tanh:** smooth features, the fit stays linear.
- **ReLU:** kinks inject curvature between points.

The cure: penalize the kinks between points



$u(x) - u^*(x)$: the fit kinks between grid points.



Regularized: flat at 10^{-15} at every width.

Penalize the interior kinks (curvature between points): the fit **snaps back** to the truth.

Conclusion

- Robust across **four models** and smooth activations.
- **The parameter count was never the risk.**